Chapter 3 — Control Flow: Decisions and Loops

Control flow is how a program chooses *which* statements run (`if`, `match`) and *how many times* (`for`, `while`). With Chapter 2's truthiness rules (empty and zero values count as false) these constructs express any algorithm.

#	Topic	Key ideas
3.1	Conditionals overview	branching and where it's used
3.2	`if` / `elif` / `else`	order of tests, guard clauses
3.3	Conditional expressions	`a if cond else b`
3.4	`match` / `case`	literal, OR, capture, wildcard, sequence, mapping, class patterns, guards, `as`
3.5	Loops and the iteration protocol	iterables, iterators, `for` vs `while`
3.6	`for` loops	dicts, nesting, `range`, `enumerate`, `zip`
3.7	`while` loops	unknown counts, infinite loops
3.8	Loop control	`break`, `continue`, `pass`
3.9	`for-else` / `while-else`	"no break happened"
3.10	Loop idioms	`sum` / `any` / `all`, counting, safe modification

Code: `code/conditionals.py`, `code/pattern_matching.py`, `code/loops.py` — outputs are real (CPython 3.14).

3.1 Conditional statements — overview

What it is

A **conditional statement** evaluates a **condition** (any expression, read by truthiness) and runs a block only when it holds. Two constructs:

- `if` / `elif` / `else` — test conditions one after another.
- `match` / `case` (3.10+) — compare one value against **patterns** describing its shape; a far more powerful cousin of C/Java `switch`.

Why it matters — real-world uses

Typical uses: input validation (reject a short password), navigation (logged in → dashboard), games (health ≤ 0 → game over), alerts (heart rate out of range), web routing by HTTP method and path (404 if none matches).

3.2 `if`, `elif`, `else`

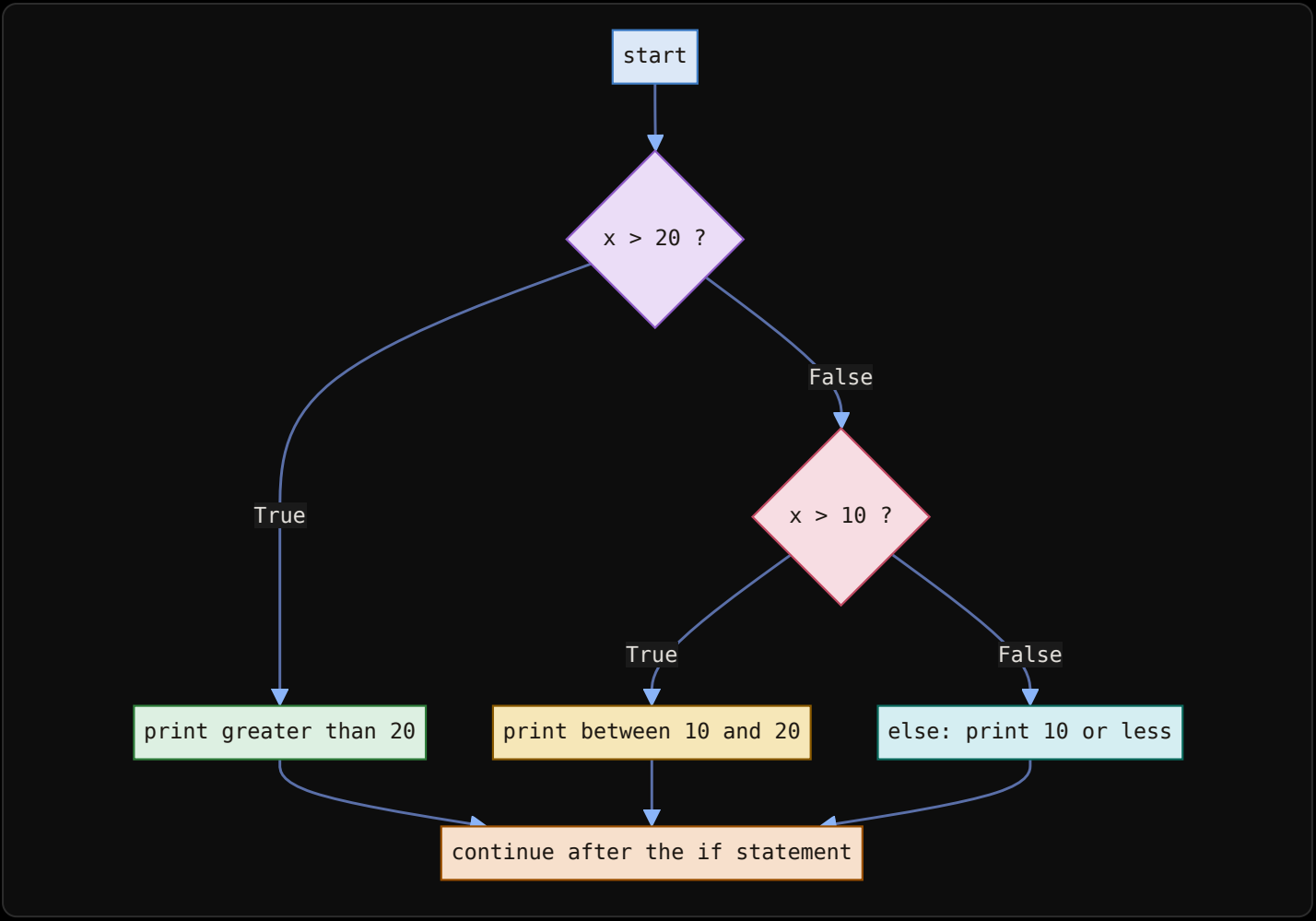
What it is

```
if condition1:
    ...            # runs if condition1 is truthy
elif condition2:
    ...            # runs if condition1 was falsy and condition2 is truthy
else:
    ...            # runs if every condition above was falsy
```

- `if` required; any number of `elif`; optional `else` last. Headers end with `:`.

- **At most one** branch runs: the first truthy one. Later conditions are not evaluated.

How it works



From `code/conditionals.py`:

```
x = 10
if x > 5:
    print("x is greater than 5")          # runs

x = 3
if x > 5:
    print("x is greater than 5")
else:
    print("x is not greater than 5")      # runs

def describe(x: int) -> str:
    if x > 20:
        return "x is greater than 20"
    elif x > 10:
        return "x is greater than 10 but not more than 20"
    else:
        return "x is 10 or less"
```

Output:

```
x is greater than 5
x is not greater than 5
25 -> x is greater than 20
15 -> x is greater than 10 but not more than 20
5 -> x is 10 or less
```

Order matters

The first true branch wins, so test the **most specific** condition first:

```python
def grade(score: int) -> str:
    if not 0 <= score <= 100:
        raise ValueError(f"score out of range: {score}")
    if score >= 90:
        return "A"
    elif score >= 80:
        return "B"
    elif score >= 70:
        return "C"
    return "F"
```

```
grades: ['A', 'B', 'C', 'F']
```

If `score >= 70` came first, every A and B student would get a C.

Nesting vs guard clauses

Deep nesting ("arrow code") is hard to read. A **guard clause** handles an edge case early with `return`/`raise` (`raise` signals an error — Chapter 9), keeping the main path flat — "flat is better than nested":

```python
def shipping_cost(total: float, is_member: bool) -> float:
    if total <= 0:
        return 0.0
    if is_member or total >= 50:
        return 0.0
    return 4.99
```

```
shipping: 4.99 0.0 0.0
```

Writing good conditions

- Truthiness: `if not cart:` rather than `if len(cart) == 0:`.
- Test `None` explicitly when `0`/`""` are valid — `discount = 0` looks "missing" to a truthiness test:

```
cart is empty
hello, guest
falsy check says missing: True | is-None check says missing: False
```

- Chained ranges: `if 18 <= temp <= 26 and not raining:`; membership instead of `or` chains: `if day in {"sat", "sun"}:`.

```
weekend
good weather for a walk
```

Trade-offs

Approach	Good for	Weak at
`if/elif/else`	Arbitrary conditions, ranges, few branches	Many equality tests on one value
`match/case`	One value, many shapes; destructuring (pulling parts into names)	Plain boolean conditions
dict lookup `{"a": f, "b": g}[key]()`	Constant keys → action (a *dispatch table*)	Conditions that aren't exact keys

Common mistakes

- `=` instead of `==` → `SyntaxError`; `else if` → must be `elif`.
- `if x == 1 or 2:` — always true; write `if x in (1, 2):`.
- `if flag == True:` → `if flag:`.
- Overlapping ranges in the wrong order (grading example).

Real-world examples

- Django views: `if request.method == "POST": … else: …`.
- Feature flags: `if flags.is_enabled("new_checkout", user): …`.

In the interview

> **30-second answer:** "`if` evaluates conditions top-down by truthiness and runs only the first matching branch; `elif` chains alternatives and `else` catches the rest. I order conditions from most specific to most general, use guard clauses to avoid deep nesting, and `is None` when zero or empty values are legitimate."

- *Does Python have `switch`?* → `match` (3.10+) covers it and more; dicts serve as dispatch tables.
- *Are later `elif`s evaluated after a match?* → No.

Self-check

1. With `x = 20`, which branch of `describe` runs? → The `elif` (`20 > 20` false, `20 > 10` true).
2. "Between 1 and 10 inclusive"? → `1 <= n <= 10`.

TL;DR

First truthy branch wins; order specific → general; flat guard clauses; use `in`, chained comparisons and truthiness.

3.3 Conditional expressions (the "ternary")

What it is

The **expression** form of `if/else` (three parts, hence "ternary"):

```
value_if_true if condition else value_if_false
```

```
parity = "even" if n % 2 == 0 else "odd"          # 7 is odd
print("Access granted." if has_access else "Access denied.")
```

Only the chosen branch is evaluated; `else` is required. Very low precedence (just above `lambda`), so parenthesise inside bigger expressions. Example: `abs` = `x if x >= 0 else -x`.

When not to use it

- Long branches or side effects (printing, changing data) — use an `if` statement.
- Nested ternaries (`a if x else b if y else c`) — legal but hard to read.
- For defaults `x or default` is shorter — only when all falsy `x` should be replaced.

TL;DR

`a if cond else b` picks a value; keep it short and un-nested.

3.4 `match` / `case` — structural pattern matching

What it is

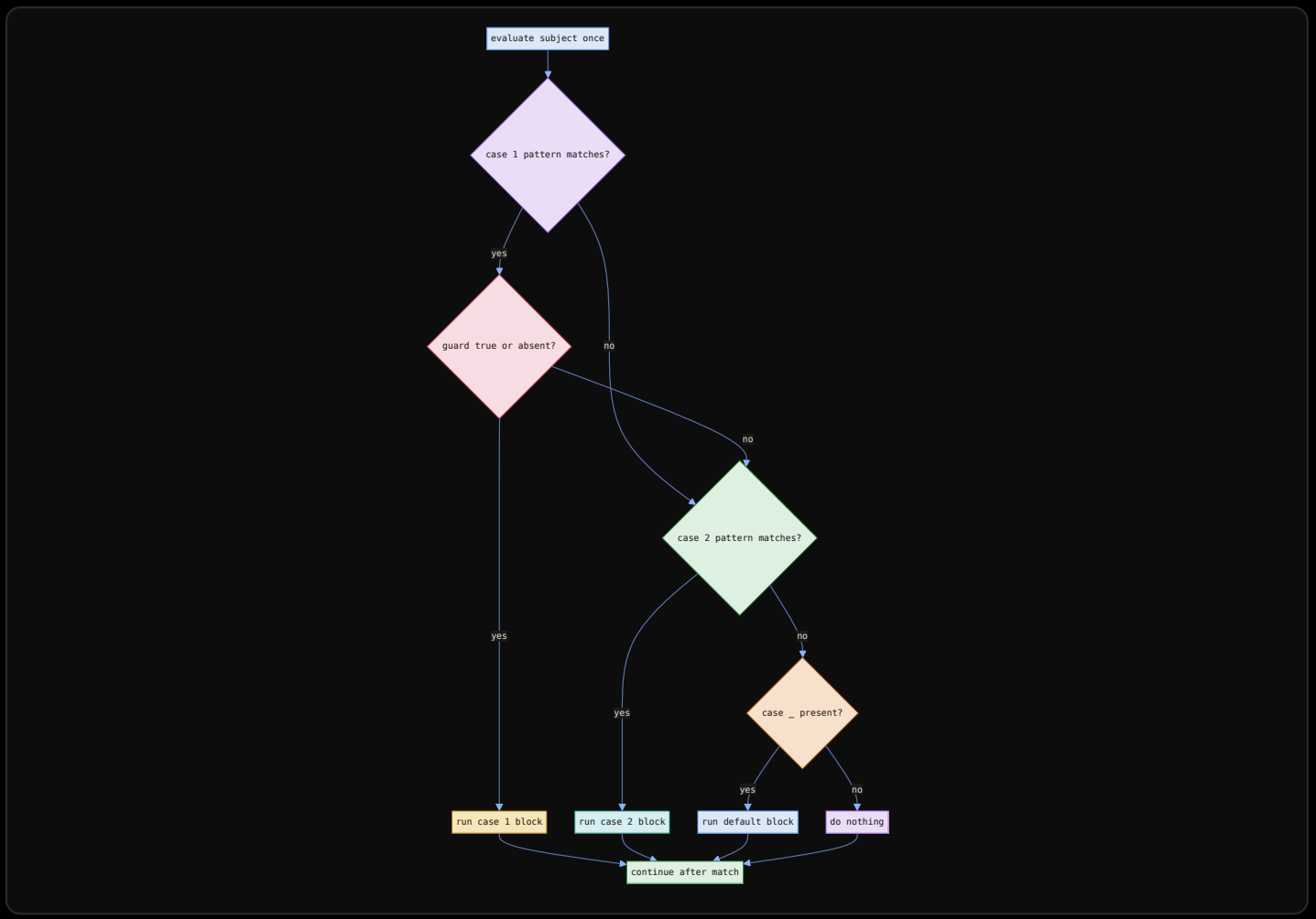
`match` (3.10, PEPs 634–636) evaluates a **subject** once and tries each `case` **pattern** in order; the first that matches (and whose optional guard is true) runs. No match → nothing happens, no error.

Unlike C/Java `switch`, a pattern describes the **structure** of the data — "a list starting with `"go"` plus one item", "a dict whose `type` is `"click"`", "a `Point` with `y == 0`" — and **binds** parts to names in one step (*destructuring*).

```
match subject:
    case pattern1:
        ...
    case pattern2 if guard:
        ...
    case _:
        ...          # wildcard: the default
```

`match`/`case` are **soft keywords** — `match` still works as a variable name.

How it works



No fall-through (C cases without `break` run into the next); you never write `break`.

The pattern kinds

Pattern	Looks like	Matches when…	Binds
Literal	`200`, `"Monday"`, `None`, `True`	subject `==` literal (`None`/`True`/`False` via `is`)	—
Capture	`x` (bare name)	always	subject → `x`
Wildcard	`_`	always	nothing
Value	`Color.RED`, `http.HTTPStatus.OK` (dotted)	subject `==` that value	—
OR	`p1 | p2 | p3`	any alternative matches	all alternatives must bind the same names
AS	`pattern as name`	pattern matches	matched part → `name`
Sequence	`[a, b]`, `(a, b)`, `[first, *rest]`	subject is a sequence (**not** `str`/`bytes`) of matching length and items	items
Mapping	`{"type": "click", "pos": p}`	mapping has those keys with matching values; **extra keys ignored**	values; `**rest` collects the others
Class	`Point(x=0, y=0)`, `Point(x, 0)`, `int(n)`, `str()`	`isinstance(subject, Point)` and listed attributes match	attributes
Guard	`case p if cond:`	pattern matches **and** `cond` truthy	(pattern's bindings)

Literal and OR patterns

From `code/pattern_matching.py`:

```python
def http_status(code: int) -> str:
    match code:
        case 200:
            return "Success"
        case 404:
            return "Not Found"
        case 500:
            return "Server Error"
        case 301 | 302 | 307 | 308:          # OR pattern
            return "Redirect"
        case _:                              # wildcard
            return "Unknown status code"

def day_kind(day: str) -> str:
    match day:
        case "Monday" | "Friday":
            return "Start/end of the work week"
        case "Tuesday" | "Wednesday" | "Thursday":
            return "Midweek"
        case _:
            return "Weekend"
```

```
['Success', 'Not Found', 'Server Error', 'Redirect', 'Unknown status code']
Midweek | Start/end of the work week | Weekend
```

Sequence patterns

```python
def first_color(colors: list[str]) -> str:
    match colors:
        case []:
            return "empty list"
        case ["red", *others]:        # first item "red"; others = the rest (a list)
            return f"red first, {len(others)} others"
        case ["blue", *_]:            # *_ = any number of remaining items, not bound
            return "blue first"
        case [single]:                # exactly one item
            return f"only one color: {single}"
        case _:
            return "different first color"
```

```
[] -> empty list
['red', 'blue', 'green'] -> red first, 2 others
['blue', 'red'] -> blue first
['pink'] -> only one color: pink
['pink', 'red'] -> different first color
```

Without a star the length must match exactly; at most one star per sequence pattern.

Command parsing, with an OR sub-pattern captured by `as` and a **guard**:

```python
def command(text: str) -> str:
    match text.split():
        case ["go", ("north" | "south" | "east" | "west") as direction]:
            return f"moving {direction}"
        case ["take", item]:
            return f"picked up {item}"
        case ["drop", *items] if items:          # guard: need at least one item
            return f"dropped {', '.join(items)}"
        case ["quit" | "exit"]:
            return "bye"
        case _:
            return f"unknown command: {text!r}"
```

```
'go north'       -> moving north
'take lamp'      -> picked up lamp
'drop key coin'  -> dropped key, coin
'drop'           -> unknown command: 'drop'
'quit'           -> bye
'dance'          -> unknown command: 'dance'
```

`"drop"` alone matches `["drop", *items]` with `items == []`, but the guard fails, so matching continues to `_`.

Tuples work the same way:

```python
match person:                                  # person = (25, "Engineer")
    case (age, "Engineer") if age > 20:
        return f"Adult Engineer aged {age}"
    case (age, profession):
        return f"{profession} aged {age}"
```

```
Adult Engineer aged 25 | Engineer aged 19
```

Strings are not sequences for matching (deliberate, so `"ab"` doesn't match `[a, b]`):

```
'ab' is matched as a str, not as a sequence of characters
```

Mapping patterns

Ideal for JSON-like events and API payloads:

```python
match event:
    case {"type": "click", "pos": (x, y)}:          # nested sequence pattern
        return f"click at {x},{y}"
    case {"type": "key", "key": str(k)} if len(k) == 1:
        return f"key {k!r}"
    case {"type": "resize", **rest}:                # rest = remaining key/values
        return f"resize with {rest}"
    case {"type": other}:
        return f"unhandled event type {other!r}"
    case _:
        return "not an event"
```

```
{'type': 'click', 'pos': (3, 4), 'button': 1} -> click at 3,4
{'type': 'key', 'key': 'q'} -> key 'q'
{'type': 'resize', 'w': 800, 'h': 600} -> resize with {'w': 800, 'h': 600}
{'type': 'scroll'} -> unhandled event type 'scroll'
['click'] -> not an event
```

The first event's extra `"button"` key doesn't prevent the match.

Class patterns

> **Key term — class:** a blueprint for objects of your own type (Chapter 10); `Point(1, 2)` is an *instance* of `Point`. `isinstance(obj, Point)` asks "was `obj` made from `Point` or a subclass?".

`ClassName(...)` checks `isinstance`, then attributes. Keyword form `Point(x=0)` matches attribute `x`. Positional form `Point(a, b)` needs `__match_args__` — a tuple naming the attributes positions map to; `@dataclass` (a decorator that auto-writes common class code) generates it (`('x', 'y')`). Built-ins like `int`, `str`, `float` take one positional sub-pattern matching the whole subject.

```python
@dataclass
class Point:
    x: float
    y: float

match shape:
    case Point(x=0, y=0):                           # keyword attributes
        return "the origin"
    case Point(x, 0):                               # positional via __match_args__
        return f"on the x-axis at {x}"
    case Point():                                   # any Point
        return "some other point"
    case Circle(center=Point(x=0, y=0), radius=r):  # nested patterns
        return f"circle of radius {r} at the origin"
    case Circle(radius=r) if r <= 0:
        return "invalid circle"
    case int(n) | float(n):
        return f"a number {n}"
    case str() as s:
        return f"a string of length {len(s)}"
    case _:
        return "unknown shape"
```

```
Point(x=0, y=0) -> the origin
Point(x=5, y=0) -> on the x-axis at 5
Point(x=1, y=2) -> some other point
Circle(center=Point(x=0, y=0), radius=2) -> circle of radius 2 at the origin
Circle(center=Point(x=1, y=1), radius=-1) -> invalid circle
7 -> a number 7
2.5 -> a number 2.5
'hello' -> a string of length 5
[1] -> unknown shape
Point.__match_args__ = ('x', 'y')
```

The capture-pattern trap

A **bare name** is a *capture*, never a comparison. With `RED = "red"`, `case RED:` matches anything and rebinds `RED`:

```
matched?! blue
SyntaxError: name capture 'RED' makes remaining patterns unreachable
```

Python refuses to compile it if more cases follow. Compare against a constant with a **dotted name** such as an `Enum` member (a named constant in an enumeration class), a literal, or a guard (`case x if x == RED:`):

```python
class Color(Enum):
    RED = "red"
    GREEN = "green"

match c:
    case Color.RED:      # compared with ==
        return "stop"
    case Color.GREEN:
        return "go"
```

```
stop go
```

Trade-offs

	`match`	`if/elif`
Check structure and extract parts at once	✅	Manual `isinstance`, `len`, indexing
Arbitrary boolean conditions	Via guards only	✅
Python version	3.10+	All
Performance	Similar — cases tried in order, not a jump table	Similar

Don't use it for one or two boolean tests (`if`), constant key → value maps (a dict), or code that must run on ≤ 3.9.

Common mistakes

- `case RED:` with a plain variable (capture trap).
- Expecting fall-through or writing `break` in a case.
- Expecting `case [a, b]:` to match a string.
- No `case _:` → unexpected input silently ignored; consider raising in the default.
- OR patterns binding different names (`case [x] | [x, y]:`) → `SyntaxError`.

Real-world examples

- CLI and chat-bot command parsers (`command()` above).
- Dispatching JSON messages from WebSockets or message queues by `type` (see System Design Fundamentals).
- Compilers walking an AST: `case BinOp(left, Add(), right):` — the `ast` node classes define `__match_args__` for this.

In the interview

30-second answer: "`match` evaluates a subject once and tries case patterns top to bottom, with no fall-through. Patterns can be literals, captures, the `_` wildcard, dotted-name constants, OR-patterns, sequences with `*rest`, mappings that ignore extra keys, and class patterns that check

> `isinstance` and attributes via `__match_args__`. Guards add an `if` condition. The main trap is that a bare name captures instead of comparing."

- *Match a constant?* → Dotted name, literal, or guard.
- *Mapping patterns need exact keys?* → No; extra keys ignored, `**rest` captures them.
- *What makes positional class patterns work?* → `__match_args__`, auto-generated by `dataclass` and `namedtuple`.

Self-check

1. What does `case [x, *_]:` match? → Any non-string sequence with ≥ 1 item; binds the first to `x`.
2. No case matches and no `_`? → Nothing; execution continues after `match`.
3. Capture which alternative of `"a" | "b"` matched? → `("a" | "b") as choice`.

TL;DR

`match` = one value against shape-aware patterns, first match wins, no fall-through; literals, `|`, `[... *rest]`, `{key: ...}`, `Class(attr=...)`, guards, `as`; dotted names for constants — bare names capture.

3.5 Loops — overview and the iteration protocol

What it is

Loop	Repeats…	Use when
`for`	once per item of an **iterable**	you have a collection or known count
`while`	while a **condition** stays truthy	the count isn't known in advance

Loop control statements: `break` (leave), `continue` (next iteration), `pass` (placeholder). Visiting every item is **traversal**.

How `for` really works — the iteration protocol

An **iterable** hands out items one at a time (lists, strings, dicts, sets, ranges, files, generators — objects producing values on demand); an **iterator** tracks the position.

1. `for x in obj:` calls `iter(obj)` to get an iterator.
2. Each round it calls `next(iterator)` and binds `x`.
3. When `next` raises `StopIteration`, the loop ends quietly.

```
it = iter(["a", "b"])
next(it)   # 'a'
next(it)   # 'b'
next(it)   # raises StopIteration
```

So one `for` works for every container, and lazy objects (`range`, file handles) are looped without loading everything into memory.

3.6 The `for` loop

What it is

```
for variable in iterable:
    ...   # body runs once per item; variable holds the current item
```

How it works — basic traversal

From `code/loops.py`:

```
for fruit in ["apple", "banana", "cherry"]:
    print(fruit, end=" ")
for ch in "hey":
    print(ch.upper(), end="")
```

```
apple banana cherry
HEY
```

No index bookkeeping, so no off-by-one errors. The loop variable **survives** the loop with its last value (unless the iterable was empty).

Looping through dictionaries

Iterating a dict yields **keys** (insertion order). Views (live read-only windows) give the rest:

You need	Write
keys	`for k in d:` (or `d.keys()`)
values	`for v in d.values():`
both	`for k, v in d.items():`

```
info = {"name": "John", "age": 30, "city": "New York"}
for key in info:
    print(key, "->", info[key])
for key, value in info.items():          # preferred: no second lookup
    print(f"{key}={value}", end=" ")
```

```
name -> John
age -> 30
city -> New York
name=John age=30 city=New York | values: ['John', 30, 'New York']
```

Nested loops

The inner loop runs fully for each outer item: `rows × cols` iterations.

```
matrix = [[1, 2, 3], [4, 5, 6]]
for row in matrix:
    for num in row:
        print(num, end=" ")
    print()                              # newline after each row
```

```
1 2 3
4 5 6
```

Two nested loops over n items are $O(n^2)$ (10× data → 100× work). If the inner loop only searches, a set/dict lookup often removes it.

`range()` — counting loops

Lazy integers, stop **exclusive**:

Call	Produces
`range(stop)`	0 … stop−1
`range(start, stop)`	start … stop−1
`range(start, stop, step)`	start, start+step, … while < stop (> stop for negative step)

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9] [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] [1, 3, 5, 7, 9] [10, 7, 4, 1]
```

`step` can't be 0 (`ValueError`). `range` supports `len()`, indexing, fast `in`, and is cheap at any size.

`enumerate()` — index and item together

Instead of `for i in range(len(names)): names[i]`:

```
for i, name in enumerate(names, start=1):
    print(f"{i}. {name}", end="  ")
```

```
1. ann  2. bob  3. cy
```

`zip()` — walk several iterables in parallel

`zip(a, b)` yields tuples `(a[0], b[0]), (a[1], b[1]), …` and **stops at the shortest**; `strict=True` (3.10+) raises if lengths differ — usually what you want for aligned data.

```
for name, score in zip(names, scores, strict=True):
    print(f"{name}:{score}", end="  ")
```

```
ann:91  bob:78  cy:85
zip strict: zip() argument 2 is shorter than argument 1
plain zip stops at the shortest: [(1, 'a'), (2, 'b')]
```

`dict(zip(keys, values))` builds a dict from two lists.

Common mistakes

- `for i in range(len(items)):` when you need items (iterate directly) or both (`enumerate`).
- `range(1, 10)` stops at 9.
- Expecting dict iteration to give pairs.
- Silent truncation with plain `zip`.

Real-world examples

- `for line in f:` streams a file; `csv.reader(f)` rows.
- `for i, batch in enumerate(dataloader):` — the standard PyTorch training loop.
- `zip(features, labels)` in ML pipelines.

In the interview

> **30-second answer:** "A `for` loop calls `iter()` on the iterable and `next()` until `StopIteration`, so it works on anything iterable. I iterate items directly, use `enumerate` for the index, `zip(strict=True)` for parallel sequences, `.items()` for dicts, and `range` for counts."

- *`zip` with unequal lengths?* → Stops at shortest; `strict=True` raises `ValueError`.
- *Loop variable after the loop?* → Keeps the last value.

Self-check

1. `list(range(2, 11, 3))`? → `[2, 5, 8]`.
2. Nested loops over 3 and 4 items: inner body runs? → 12 times.

TL;DR

`for item in iterable`; dict → keys (use `.items()`); `range` excludes stop; `enumerate` for indices; `zip(strict=True)` for parallel lists.

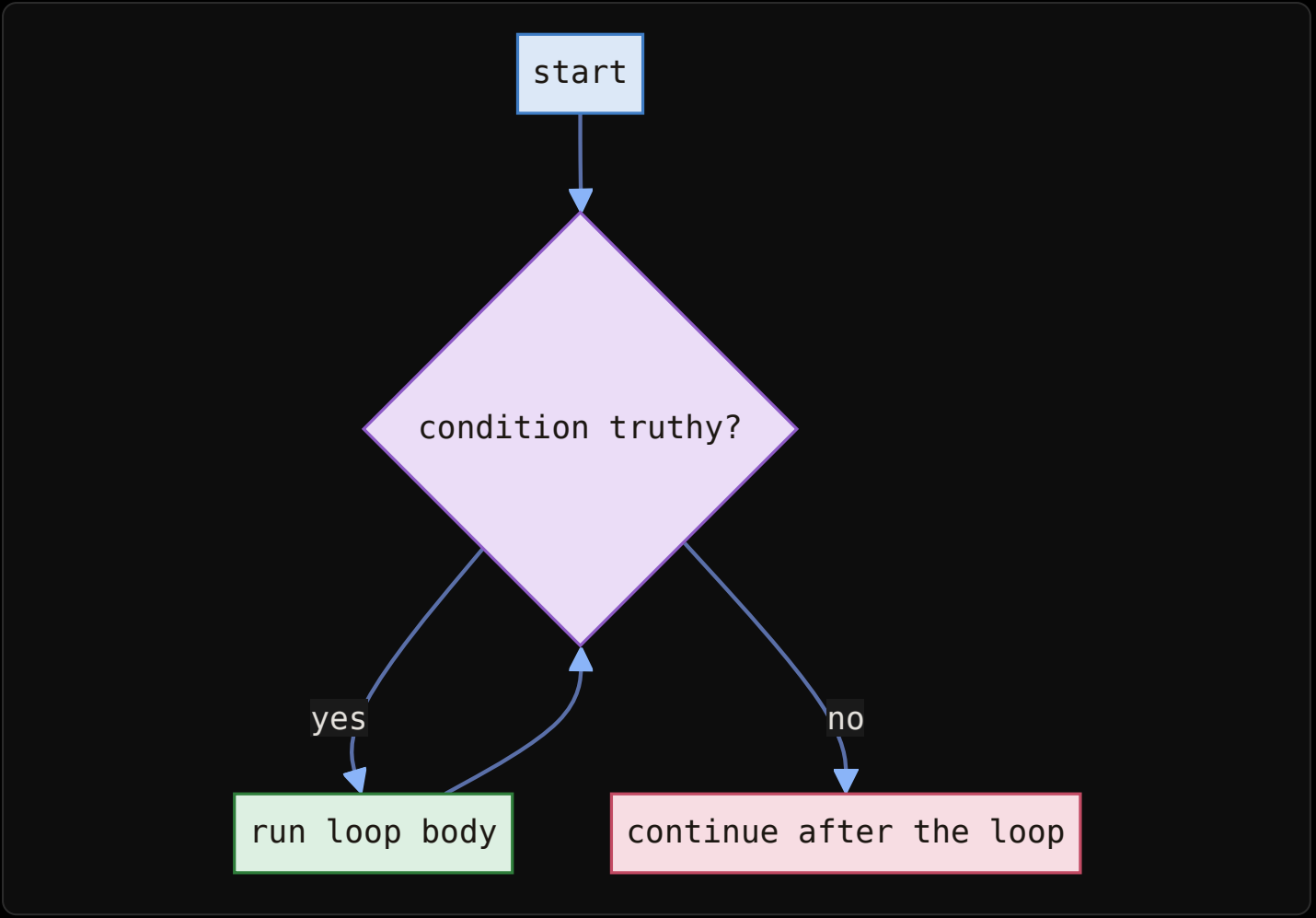
3.7 The `while` loop

What it is

```
while condition:
    ...   # repeats while condition is truthy; re-checked before every iteration
```

For unknown iteration counts: read until "quit", retry until success, iterate until a value converges.

How it works



The condition is checked first (the body may run **zero** times), and the body **must** change something the condition depends on.

```
count = 0
while count < 5:
    print(f"Count is {count}", end="; ")
    count += 1          # progress toward making the condition false
```

```
Count is 0; Count is 1; Count is 2; Count is 3; Count is 4;
```

A genuinely unknown count — halvings until below 1:

```
n, steps = 100.0, 0
while n >= 1:
    n /= 2
    steps += 1
```

```
halvings until < 1: 7
```

(100 → 50 → 25 → 12.5 → 6.25 → 3.125 → 1.5625 → 0.78125.)

Infinite loops on purpose

`while True:` + `break` is the idiomatic "loop and a half" (menus, server loops):

```
while True:
    answer = input("> ")
    if answer == "quit":
        break
    handle(answer)
```

With walrus: `while (answer := input("> ")) != "quit": handle(answer)`.

Trade-offs

	`for`	`while`
Termination	Automatic when exhausted	Your responsibility
Typical bug	Off-by-one in `range`	Infinite loop (forgot to update)
Use for	Collections, fixed counts	Unknown counts, event loops, retries, convergence

Common mistakes

- Forgetting the update → infinite loop (Ctrl-C raises `KeyboardInterrupt`).
- `while i < len(items): … i += 1` to traverse — use `for`.
- Float conditions like `while x != 1.0:` that never become exactly true — use `<` / `>` or a tolerance.

Real-world examples

- Game loops: `while running: handle_events(); update(); draw()`.
- Retry with back-off: `while attempts < 5 and not ok: …`.
- Chunked reading: `while (chunk := f.read(8192)): …`.

In the interview

> **30-second answer:** "`while` re-checks its condition before each iteration and is for unknown iteration counts. The body must make progress toward the condition becoming false; for intentional infinite loops I use `while True` with a clear `break`."

TL;DR

`while cond:` repeats until `cond` is falsy (possibly zero times); guarantee progress; `while True` + `break` for loop-and-a-half.

3.8 Loop control: `break`, `continue`, `pass`

What it is

Statement	Effect
`break`	Exit the **innermost** loop now; skips any loop `else`
`continue`	Skip to the next iteration (`while` re-checks its condition)
`pass`	Do nothing; fills a place where a statement is required

How it works

```python
for number in [1, 2, 3, 4, 5, 6]:
    if number == 4:
        print("Number 4 found, stopping loop.")
        break                                   # 5 and 6 are never visited

odds = []
for number in [1, 2, 3, 4, 5, 6]:
    if number % 2 == 0:
        continue                                # skip evens
    odds.append(number)

for number in [1, 2, 3]:
    if number % 2 == 0:
        pass                                    # placeholder - does nothing
    print(number, end=" ")                      # still runs for every number
```

```
Number 4 found, stopping loop.
continue kept odds: [1, 3, 5]
1 2 3
```

`pass` is **not** `continue`: the rest of the body still runs. Use it (or `...`) for stubs: `def todo(): pass`, `class MyError(Exception): pass`.

`break` exits only one level:

```python
for i in range(3):
    for j in range(3):
        if j == 1:
            break                   # leaves the inner loop only
        print(f"({i},{j})", end=" ")
```

```
(0,0) (1,0) (2,0)
```

To exit several levels, move the loops into a function and `return`, or use a flag variable.

Common mistakes

- Expecting `break` to leave all nested loops.
- Confusing `pass` and `continue`.
- `continue` in a `while` *before* the counter update → infinite loop.
- `continue`-based filtering where a comprehension is clearer.

In the interview

> **30-second answer:** "`break` leaves the innermost loop, `continue` jumps to the next iteration, `pass` is a no-op placeholder. To break out of nested loops I refactor into a function and `return`."

TL;DR

`break` = leave loop, `continue` = next iteration, `pass` = do nothing; all act on the innermost loop.

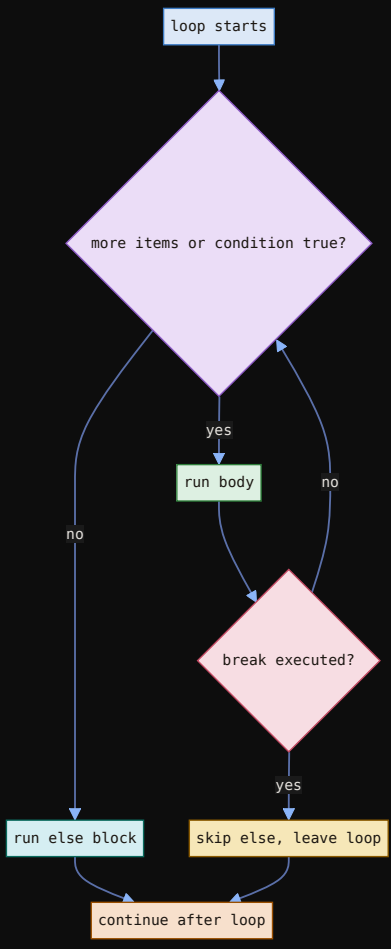
3.9 `for-else` and `while-else`

What it is

A loop's `else` runs **only if the loop finished without** `break` (iterable exhausted, or condition became false). Read it as **"no break"**. Classic use: search loops — `break` = found, `else` = not found, no flag needed.

```python
for item in iterable:
    ...
else:
    ...   # runs only if the loop didn't break
```

How it works



From `code/loops.py`:

```python
for number in [1, 2, 3, 4, 5]:
    if number == 3:
        print("Number 3 found, stopping search.")
        break
else:
    print("Number not found.")        # skipped: break happened

for number in [1, 2, 4, 5]:
    if number == 3:
        print("found 3")
        break
else:
    print("3 not in list (else ran: no break)")

count = 0
while count < 5:
    if count == 2:
        print("Breaking at count 2.")
        break
    count += 1
else:
    print("Loop was uninterrupted.")  # skipped: break happened
```

```
Number 3 found, stopping search.
3 not in list (else ran: no break)
Breaking at count 2.
```

With no `break` in the body the `else` always runs — legal but pointless.

Inside a function, `return` plays `break` and the final statement plays `else` — e.g. primality by trial division (try every divisor up to √n):

```python
def is_prime(n: int) -> bool:
    if n < 2:
        return False
    for d in range(2, int(n ** 0.5) + 1):
        if n % d == 0:
            return False
    return True
```

```
is 29 prime? True | is 21 prime? False
```

At module level, or when you need to continue afterwards, `for-else` is the flag-free equivalent.

Trade-offs / when not to use it

Many developers misread `else` as "if the loop didn't run". If that confuses your team, use a helper with `return`, or `any()` / `next()` (§3.10). Comment it `# no break`.

Common mistakes

- Thinking `else` runs only when the body never executed — it also runs after a normal full run.
- Thinking `else` runs after `break`.
- Misaligning `else` so it pairs with an `if` inside the loop (or vice versa).

In the interview

> **30-second answer:** "A loop's `else` runs only when the loop wasn't terminated by `break` — think 'no break'. It's handy for search loops: `break` on found, `else` for not-found, without a flag variable."

- *Runs on an empty iterable?* → Yes, no break happened.
- *`return` inside the loop?* → Skips `else` (leaves the function).

TL;DR

Loop `else` runs when the loop ends naturally, not via `break`; use it for search-with-not-found and comment it.

3.10 Common loop idioms

Aggregation with built-ins

Many loops are built-ins in disguise — shorter, clearer and run in C:

```python
nums = [4, -1, 7, 0, 3]
sum(nums), min(nums), max(nums)
any(x < 0 for x in nums)                # True if at least one; stops early
all(x < 10 for x in nums)               # True if every one; stops early
[x * x for x in nums if x > 0]          # list comprehension: map + filter
list(reversed(nums)), sorted(nums, reverse=True)
next((x for x in nums if x > 5), None)  # first match or a default
```

```
sum/min/max: 13 -1 7
any negative? True | all < 10? True
squares of positives: [16, 49, 9]
reversed: [3, 0, 7, -1, 4] | sorted desc: [7, 4, 3, 0, -1]
first > 5: 7
```

`next(generator, default)` replaces a whole `for … break … else` search. Comprehensions are covered in the lists chapter. Note `all([])` is `True`, `any([])` is `False`.

Counting / grouping with a dict

```
counts = {}
for word in "the cat and the hat".split():
    counts[word] = counts.get(word, 0) + 1
```

```
word counts: {'the': 2, 'cat': 1, 'and': 1, 'hat': 1}
```

(`collections.Counter` does this in one call — dictionaries chapter.)

Don't change a collection while looping over it

Removing during iteration shifts later items left, so the loop **skips** the next one:

```
buggy = [1, 2, 2, 3]
for x in buggy:
    if x == 2:
        buggy.remove(x)        # the second 2 slides into the visited slot and is skipped
```

```
remove-while-iterating left: [1, 2, 3] | filtered correctly: [1, 3]
RuntimeError: dictionary changed size during iteration
```

Fix: rebuild (`[x for x in data if x != 2]`) or iterate a copy (`for x in data[:]:`). Dicts refuse outright with `RuntimeError`; iterate `list(d)` if you must add/delete keys.

Idiom cheat-sheet

Instead of…	Write…
`for i in range(len(xs)): print(xs[i])`	`for x in xs: print(x)`
index + item via `range(len())`	`for i, x in enumerate(xs):`
parallel indexing `a[i], b[i]`	`for x, y in zip(a, b, strict=True):`
`for k in d: v = d[k]`	`for k, v in d.items():`
looping backwards with indices	`for x in reversed(xs):`
accumulator loop (keeps a running total)	`sum(xs)`
flag + loop to test a condition	`any(...)` / `all(...)`
loop with `append` to transform	list comprehension
`while` loop reading chunks	`while (chunk := f.read(n)):`

Real-world examples

- Log scanning: `for line in open(path): if "ERROR" in line: …`.
- ETL jobs (extract–transform–load) aggregating per-key totals in dicts.
- Validation: `if not all(field in payload for field in REQUIRED): return 400`.

In the interview

> **30-second answer:** "I replace hand-written loops with built-ins where possible — `sum`, `any`, `all`, `min` / `max`, `sorted`, comprehensions, `next` with a default — because they're clearer and run in C. I never mutate a list or dict while iterating it; I build a new one or iterate a copy."

- *Why does removing during iteration skip items?* → The list iterator advances by index; removal shifts later items down.
- *Do `any` / `all` evaluate everything?* → No, they short-circuit.

TL;DR

Prefer built-ins and comprehensions over manual loops; `enumerate` / `zip` / `.items()` over indexes; never mutate what you're iterating.

Chapter 3 TL;DR

- `if/elif/else`: first truthy branch wins; order specific → general; guard clauses keep code flat; `a if c else b` for values.
- `match/case` (3.10+): literals, `|`, captures, `_`, dotted constants, sequences with `*rest`, mappings (extra keys ignored), class patterns (`__match_args__`), guards, `as`; no fall-through; bare names capture.
- `for` uses the iteration protocol (`iter` / `next` / `StopIteration`); dicts yield keys; use `range`, `enumerate`, `zip(strict=True)`, `.items()`.
- `while` runs while a condition holds — guarantee progress; `while True` + `break` for loop-and-a-half.
- `break` leaves the innermost loop, `continue` skips ahead, `pass` does nothing.
- Loop `else` = "no break happened" — ideal for search loops.
- Reach for `sum` / `any` / `all` / `next` /comprehensions first; never modify a collection while iterating it.